

THOMAS M

Durgapur, West Bengal, India

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Summary

Computer Science Engineer pursuing M.Tech in Operations Research at NIT Durgapur. Interested in backend development, algorithms, and optimization, with experience in Java, Spring Boot, Android development, and machine learning.

Education

Sri Sairam Institute of Technology

B.E. Computer Science and Engineering — CGPA: 8.53/10

2022 – 2026

Chennai, Tamil Nadu

National Institute of Technology Durgapur

M.Tech, Operations Research

2026 – 2028

Durgapur, West Bengal

Experience

Hexaind Technologies and Private Ltd

Jun 2024 – Sep 2024

AI/ML Intern (Remote)

- Spearheaded a real-time face and eye detection pipeline utilizing OpenCV vision frameworks.
- Processed and engineered a dataset of **500+ image samples** to fine-tune model reliability.
- Implemented targeted algorithmic optimizations to reduce frame processing latency by **15%**.
- Integrated individual modules into **1 cohesive system** engineered for automated vision tasks.

Projects

Airline Reservation System | *Java, Spring Boot, MySQL, Hibernate*

- Established a modular Airline Reservation System with **10+ integrated backend modules** using Spring Boot.
- Orchestrated **6 end-to-end workflows** including booking, payment, ticketing, cancellation, and notifications.
- Delivered **30+ REST APIs** with JPA, validation, exception handling, and Swagger documentation.

TravelAssist – Roadside Assistance Platform | *Java, Android Studio, Spring Boot, MongoDB*

- Launched an Android application with **3 real-time roadside assistance services**.
- Connected **10+ Spring Boot REST APIs** with the Android application.
- Maintained **5+ MongoDB collections** for users, requests, and assistance records.

QuantMatch – Electronic Trading Platform | *Java, Spring Boot, MySQL, Redis*

- Constructed a backend trading platform implementing a Limit Order Book using Price-Time Priority.
- Exposed **6+ REST APIs** for order placement, matching, execution, and management.
- Optimized order book retrieval using Redis caching to improve system performance.

Technical Skills

Languages: Java, C++, C, Python

Backend: Spring Boot, REST APIs, JPA, Hibernate

Databases: MongoDB, MySQL, Firebase

Mobile Development: Android Studio, XML

Core Computer Science: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, Postman

Achievements

- Qualified GATE Computer Science and Information Technology (CS) examination twice.
- Solved 440+ Data Structures and Algorithms problems on LeetCode.

Certifications

NPTEL: Cloud Computing, Internet of Things, Python for Data Science, Software Conceptual Design

MathWorks: Machine Learning Onramp, Deep Learning Onramp

MongoDB: Introduction to MongoDB, AI and Vector Search

Coursera: Business Analysis and Process Management